

ORIGINAL ARTICLE OPEN ACCESS

Informal Digital Learning of English (IDLE) as Form-Focused and Meaning-Focused Activities: Refining its Measurement and Examining its Predictive Role in L2 Achievement and Confidence

Minlin Zou¹  | Mark Feng Teng²  | Ali Soyoo³  | Xueyun He⁴

¹School of Education, University of Exeter, Exeter, UK | ²Faculty of Languages and Translation, Macao Polytechnic University, Macau, China | ³Department of English Language Education, The Education University of Hong Kong, Hong Kong SAR, China | ⁴School of Foreign Languages, Hengyang Normal University, Hengyang, China

Correspondence: Minlin Zou (mz405@exeter.ac.uk)

Received: 23 February 2025 | **Revised:** 16 April 2025 | **Accepted:** 1 May 2025

Funding: This research is funded by a PhD scholarship awarded by University of Exeter and the China Scholarship Council of the Ministry of Education of China [grant number CSC. NO. 202308430033].

Keywords: form-focused IDLE activities | informal digital learning of English (IDLE) | L2 confidence | meaning-focused IDLE activities | scale development and validation

关键词: 非正式数字化英语学习 (IDLE) | 形式聚焦IDLE活动 | 意义聚焦IDLE活动 | 量表开发与验证 | 二语自信

ABSTRACT

Acknowledging the limitations of existing measurement instruments in adequately capturing the distinct cognitive processes (e.g., form-focused vs. meaning-focused) involved in informal digital learning of English (IDLE), this paper asserts a pressing need to develop and validate robust instruments that measure form-focused and meaning-focused IDLE (FM-IDLE). It comprises two interrelated studies investigating the measurement and impact of informal digital English learning practices among Chinese university students. Study 1 aimed to develop and validate a psychometrically sound instrument to assess learners' participation in form-focused IDLE Activities (FIA) and meaning-focused IDLE activities (MIA). Study 2 examined the predictive relationship between FIA, MIA, and two critical language learning outcomes: L2 achievement as indicated by students' standardized English test scores and L2 confidence. The research engaged 1,202 Chinese undergraduate EFL learners, with the sample randomly divided into two datasets (Sample 1: $N = 601$; Sample 2: $N = 601$) for scale validation, while the complete sample ($N = 1202$) was utilized to examine the scale's predictive validity. Study 1 yielded a refined 14-item FM-IDLE Scale comprising two distinct factors, FIA and MIA, demonstrating robust psychometric properties. Study 2, employing structural equation modeling, revealed differential effects: FIA demonstrated significant and negative associations with L2 achievement and L2 confidence, whereas MIA could positively predict these outcomes. These findings yield important pedagogical implications by highlighting the importance of promoting meaning-focused IDLE activities through technological affordances while fostering language learning confidence against the backdrop of the global spread of English.

This is an open access article under the terms of the [Creative Commons Attribution](https://creativecommons.org/licenses/by/4.0/) License, which permits use, distribution and reproduction in any medium, provided the original work is properly cited.

© 2025 The Author(s). *International Journal of Applied Linguistics* published by John Wiley & Sons Ltd.

摘要

本研究基于现有测量工具在有效捕捉非正式数字化英语学习 (IDLE) 中“形式聚焦”与“意义聚焦”认知过程方面存在的局限性, 提出亟需开发并验证具有良好信效度的“形式聚焦”与“意义聚焦”IDLE (FM-IDLE) 测量工具。研究通过两项子研究系统考察中国大学生数字化英语学习实践的测量维度及其影响机制。研究一聚焦于开发并验证测量“形式聚焦” (FIA) 与“意义聚焦” (MIA) IDLE活动的心理测量学工具; 研究二则探究FIA与MIA对二语学业成就 (标准化英语测试成绩) 与二语自信的预测效应。研究对象为1,202名中国本科EFL学习者, 通过随机分半法将样本划分为两组 (样本1: $N = 601$; 样本2: $N = 601$) 进行量表验证, 整合全样本 ($N = 1202$) 检验量表的预测效度。研究一最终确立包含14个题项的FM-IDLE量表, 该工具具有清晰的双因子结构 (FIA与MIA), 并展现出良好的心理测量学特性。研究二采用结构方程模型揭示差异化预测效应: “形式聚焦”活动 (FIA) 与二语学业成就及自信呈显著负向关联, 而“意义聚焦”活动 (MIA) 则对此类学习结果具有显著正向预测作用。研究发现具有重要教学启示: 强调了在英语全球普及的背景下, 通过技术赋能来促进“以意义为中心”的IDLE活动, 同时培养语言学习自信的重要性。

1 | Introduction

The unprecedented pace of globalization and technological innovation has fundamentally reshaped the field of second language (L2) education, particularly in how learners engage with English beyond traditional classroom boundaries. As digital technologies become increasingly ubiquitous, informal digital learning of English (IDLE) has emerged as a significant phenomenon in computer-assisted language learning (CALL). IDLE represents “self-directed, self-guided, informal English learning by leveraging an array of digital devices (e.g., mobile phones, laptops, desktop computers) and online resources (e.g., webpages, social media platforms) detached from formal instruction style” (Lee 2019, 768). This shift toward digitally mediated informal learning marks a fundamental transformation in how learners access, process, and internalize language input in a world where online exposure to English has become more accessible and frequent than before (Sundqvist 2024).

A growing body of empirical evidence has demonstrated the multifaceted impact of IDLE practices on L2 acquisition. Informal digital learning experiences have been found to influence various dimensions of language development, including linguistic competencies, such as vocabulary acquisition (Lai and Wang 2024; Lee and Dressman 2018; Rezai et al. 2024), writing proficiency (Kaatari et al. 2023), and oral communication skills (Sundqvist and Uztosun 2024). Additionally, IDLE has been linked to psychological factors like L2 enjoyment (Lee and Lee 2020; Liu et al. 2024b) and self-efficacy (Zadorozhnyy and Lee 2023); motivational constructs (Liu 2024a; Liu and Ma 2024; Liu et al. 2024; Zou et al. 2025c); communicative behaviors (Lee and Taylor 2024); and affective regulation, particularly in reducing L2 anxiety, boredom, and burnout (Ghasemi and Azari Noughabi 2024).

While these findings have substantially enriched our understanding of how informal digital English learning contributes to L2 attainment, a critical methodological limitation persists in the field: current measurement instruments lack the capacity to differentiate between form-focused and meaning-focused digital learning activities. This distinction between form and meaning focus is theoretically significant, as these modalities engage distinct cognitive processes and potentially yield different learning outcomes (Ellis 2001; Svalberg 2021). Form-focused activities necessitate explicit attention to linguistic structures and

forms, whereas meaning-focused activities prioritize authentic communication and content comprehension. Common digital form-focused activities include grammar practice and vocabulary drilling, while meaning-focused activities engage with authentic content through social media, videos, and online discussions (Soyoo 2023; Zhang and Liu 2022). Although existing IDLE studies acknowledge both form-focused and meaning-focused practices (Lee and Draji 2019; Liu et al. 2024a; Zadorozhnyy and Lee 2023), they typically operationalize IDLE as a unitary and single variable. This methodological oversimplification potentially obscures our understanding of how different types of informal digital English learning activities distinctly contribute to L2 development, potentially leading to imprecise or misleading conclusions about the effectiveness of IDLE across contexts.

As such, the present article aims to address this methodological limitation through two interrelated objectives. First, we aim to develop and validate a new IDLE scale that explicitly distinguishes between form-focused IDLE activities (FIA) and meaning-focused IDLE activities (MIA), thus providing a more nuanced way of measuring and operationalizing IDLE. Second, we seek to examine the differential effects of these two types of IDLE activities on L2 achievement and foreign language confidence using a structural equation modeling (SEM) approach (Collier 2020; Kline 2015). The research questions (RQs) that guide the present paper include:

- RQ1:** To what extent, if any, is the form-focused and meaning-focused FM-IDLES developed reliable and valid?
- RQ2:** To what extent, if any, do FIA and MIA predict L2 English achievement?
- RQ3:** To what extent, if any, do FIA and MIA contribute to L2 confidence?

2 | Literature Review

The following section establishes the conceptual and empirical backgrounds for this study by reviewing literature across three key areas: the conceptual underpinnings and significance of IDLE in CALL, existing measurement approaches for IDLE and related constructs, and empirical evidence linking IDLE to L2 English achievement and L2 confidence.

2.1 | The Conceptual Foundation and Significance of IDLE in CALL

As Sundqvist (2024) notes, the growing prevalence of informal language learning signals a fundamental structural change in L2 education, where traditional top-down classroom instruction is increasingly complemented by bottom-up, learner-initiated approaches. Within this shift, IDLE has emerged as a significant phenomenon in L2 research, reflecting the ubiquitous role of digital technologies in providing learners with unprecedented access to language learning resources. Theoretically, IDLE is anchored at the intersection of Benson and his colleagues' four-dimensional model of informal language learning (Benson 2011; Reinders and Benson 2017) and research on digital literacies in language learning (Chik 2019; Jones and Hafner 2021; Thorne 2014).

IDLE encompasses six conceptual dimensions that capture both its informal and digital characteristics: location, formality, pedagogy, locus of control, trajectory, and digital mediation. The first four dimensions, derived from Benson's framework, establish IDLE's informal learning nature by situating it outside of classroom settings, with informal structure, self-instructed pedagogy, and learner-controlled process. The trajectory dimension (Chik 2019) acknowledges the temporal patterns of digital engagement from sustained to sporadic, while digital mediation (Jones and Hafner 2021; Thorne 2014) addresses how technological affordances shape these learning experiences. Together, these multidimensions provide a comprehensive foundation for understanding IDLE as a distinct form of language learning, particularly relevant in today's digital age, where the boundaries between formal and informal learning increasingly blur.

The autonomous and variable nature of IDLE, particularly reflected in its locus of control and trajectory dimensions, positions it as an emerging individual difference (ID) variable in L2 development (Liu et al. 2024; Soyoo et al. 2023; Sundqvist 2024). While traditional ID research has focused on internal learner characteristics such as cognitive, conative, affective, and personality factors (Skehan 1989, 1991), the increasingly technology-mediated reality of language learning necessitates incorporating learners' participation in informal digital learning as a crucial ID variable (Sundqvist 2024). This theoretical positioning is supported by extensive empirical evidence, with studies demonstrating IDLE's bidirectional relationships with other ID variables like linguistic confidence (Lee and Dressman 2018), motivation (Lee and Lu 2023; Liu et al. 2024b; Zou et al. 2025c), self-efficacy (Lee and Drajiati 2019; Zadorozhnyy and Lee 2023), enjoyment (Lee et al. 2021; Liu et al. 2024), and intercultural competence (Lee and Lee 2020; Liu et al. 2023). The systematic variations in how learners participate in IDLE activities, coupled with its demonstrable influence on multiple aspects of L2 development, establish IDLE as a distinct and critical individual difference variable in contemporary L2 development shaped by the ubiquity of digital technology.

2.2 | Research on the Measurement of IDLE or its Associated Concepts

Research efforts to measure IDLE and its associated concepts have employed various approaches, reflecting the complex nature

of informal digital English learning. A systematic review of empirical studies reveals that Likert scales have been predominantly used to assess learners' engagement with informal English learning activities beyond the classroom. For instance, Lee and Drajiati (2019) developed and operationalized IDLE through a 13-item scale distinguishing between receptive IDLE activities (RIA) and productive IDLE activities (PIA), although their scale development and validation processes were not extensively documented. In a more recent study, Liu et al. (2024a) introduced a six-item scale focusing on the use of generative artificial intelligence (AI) tools for IDLE purposes, building on the technology acceptance model. However, the narrow focus on AI tool usage may not fully capture the broader spectrum of IDLE activities that learners engage with in their daily lives.

Several scholars have developed alternative measurement approaches under the broader umbrella of informal language learning. For example, Lai et al. (2015) adopted a mixed-methods design to examine Chinese EFL secondary students' out-of-class language learning quality, incorporating both quantitative measures of activity frequency and qualitative assessments of learning patterns. Similarly, Lamb and Arisandy (2020) developed a 27-item scale measuring online informal learning of English (OILE) across three dimensions: self-instruction, entertainment, and social orientation, using a 6-point Likert scale to assess behavioral representativeness. In a more comprehensive effort, Sundqvist and Uztosun (2024) developed and validated the Extramural English (EE) scale through a large-scale study involving 906 EFL secondary students across four countries. Their scale encompasses eight dimensions with 32 items, measuring activities ranging from gaming and digital creation to social interaction and traditional literacy practices, all rated on a 5-point frequency scale (never to always). This multidimensional approach reflects the diverse ways learners engage with English in digital environments.

Drawing upon previous IDLE-related concepts (e.g., extramural English (EE) and OILE), our study aims to develop a new scale (i.e., FM-IDLES) that attends to both FIA and MIA. We elaborate on theoretical and empirical novelties of FM-IDLE that differentiate it from EE and OILE concepts. Theoretically, while existing scales like EE (Sundqvist and Uztosun 2024) and OILE (Lamb and Arisandy 2020) have advanced the measurement of informal language learning activities, current approaches predominantly focus on frequency measures rather than the quality of engagement, particularly without distinguishing form-focused and meaning-focused activities in learners' out-of-class learning practices. The form-focused and meaning-focused IDLE dichotomy is theoretically significant, as these two types of activities engage different cognitive processes (Ellis 2001; Svalberg 2021), thus potentially leading to distinct learning outcomes in informal digital contexts. In addition, the FM-IDLES is designed to capture the unique affordances and characteristics of digital learning environments. While the EE scale (Sundqvist and Uztosun 2024) includes some digital activities, it also encompasses traditional literacy practices and offline social interaction. By contrast, the FM-IDLES focuses on digital learning activities, reflecting the growing importance of technology-mediated language learning in the digital era (Reinders and Benson 2017). Empirically, the FM-IDLES employs a 6-point frequency scale, which offers a more granular assessment of learners' engagement

compared to the 5-point scale used in the EE (Sundqvist and Uzto-sun 2024). Moreover, distinguishing between these two forms of engagement could provide more nuanced insights into how different types of digital learning activities contribute to various aspects of L2 development, thereby advancing our understanding of the complex relationship between IDLE and language learning outcomes. Thus, our study aims to develop and validate an FM-IDLES that measures both FIA and MIA. To facilitate a clear understanding of the two aspects of IDLE, we provide the following definitions:

FIA refers to out-of-class and technology-mediated English learning activities that primarily target the formal aspects of language, such as grammar, vocabulary, pronunciation, and sentence structure. The main goal of FIA is to enhance learners' understanding and mastery of the linguistic forms and structures of the English language.

MIA involves out-of-class and technology-mediated English learning that emphasizes comprehension, communication, and the practical application of language in real-life contexts. The main goal of MIA is to develop learners' ability to understand and convey meaning effectively in English, focusing on tasks that promote communication and language use in authentic situations.

2.3 | Theoretical Framework

The conceptualizations of FM-IDLES encompass both form-focused and meaning-focused activities, as any type of language skills acquisition involves elements of FIA and MIA in the learning process (Lai et al. 2015; Zou et al. 2025b, 2025c). While this theoretical distinction offers a valuable framework for understanding different types of language learning activities, it is crucial to recognize that the boundaries between form-focused and meaning-focused activities are often blurred in authentic language use, particularly within informal language learning contexts (Ellis 2001). To better account for this conceptual hybridization, the present study draws upon usage-based and sociocultural perspectives on language learning. Usage-based theory emphasizes that language learning is driven by learners' exposure to and engagement with authentic linguistic input, constituting a bottom-up process (Ellis and Wulff 2020). Experiencing repetitive encounters or drills with linguistic forms in meaningful communication environments gradually enables learners to develop form-focused and meaning-focused linguistic knowledge (Ellis and Wulff 2020). When it comes to sociocultural theory, it underscores the co-construction and integration of form-focused and meaning-focused activities in a social and contextual nature (Lantolf and Thorne 2006). Situated in informal digital learning environments, learners may engage in interactive and communicative activities, which facilitate the negotiation of meaning with other users and foster the joint development of linguistic and communicative competence (Chik 2019; Liu 2025). Integrating these usage-based and sociocultural theories into the theoretical framework in this study provides a more nuanced comprehension of form-focused and meaning-focused activities.

2.4 | Empirical Studies on the Relationships Between IDLE and L2 Achievement/Confidence

Empirical research examining the relationship between informal language learning and L2 development has highlighted the benefits of engaging students in IDLE, especially in terms of how participation in IDLE enables improved L2 achievement (Lai et al. 2015; Lee and Dressman 2018; Sundqvist and Uzto-sun 2024) and strengthened L2 learning confidence (Liu and Darwin 2024). Regarding L2 achievement, several studies have demonstrated the positive impact of informal digital learning activities, although few have specifically distinguished between form-focused and meaning-focused engagement. For example, Sundqvist and Uzto-sun (2024), investigating the predictive role of EE—a construct parallel to IDLE—found positive effects on perceived speaking competence across both Scandinavian ($n = 197$) and Asian ($n = 125$) EFL contexts. Similarly, Peters et al.'s (2019) study of 138 secondary and university students revealed that involvement with online activities positively influenced English vocabulary proficiency. In a more comprehensive investigation combining quantitative and qualitative methods, Lee and Dressman (2018) demonstrated that IDLE activities, including both form-focused and meaning-focused dimensions, significantly predicted L2 achievement. Based on these consistent findings suggesting a robust relationship between IDLE and L2 achievement, we propose:

Hypothesis 1. *FIA has a positive influence on L2 achievement.*

Hypothesis 2. *MIA has a positive influence on L2 achievement.*

Beyond linguistic outcomes, IDLE's relationship with affective variables, particularly L2 confidence, merits a more thorough investigation. Traditional conceptualizations of L2 confidence as mere beliefs in one's linguistic competence have proven inadequate in today's globalized, multilingual context. Addressing this limitation, Liu (2024b) reconceptualized foreign language confidence through the lens of World Englishes. He posited that L2 confidence in an era of unprecedented mobility and linguistic diversity encompasses three interconnected dimensions: learners' assurance in their English linguistic competency, their awareness of language variation, and, very importantly, "a sense of linguistic ownership and legitimacy in the face of English as a decentered language" (Liu 2024b, 6). This expanded conceptualization of confidence as a critical affective trait carries particular significance in contemporary English language learning, where the target language increasingly functions as a global lingua franca (Lee and Lee 2020). While direct empirical examination of the relationship between IDLE activities and L2 confidence remains limited, emerging research offers promising insights. For instance, Lee and Dradjati's (2019) investigation of 183 Indonesian university EFL learners revealed significant correlations between productive IDLE activities, L2 confidence, and willingness to communicate. Through an ethnographic case study, Liu and Darwin (2024) reported two rural Chinese EFL learners who developed creative and productive IDLE trajectories. By negotiating cultural capital and resources through investing in a range of form-focused and meaning-focused literacy practices in transnational online spaces, it was found that the two learners could construct their identity as active and competent intercultural

TABLE 1 | Research design of two sub-studies.

Study	Procedure	Sample	Instrument	Data analysis
Study 1	Scale development	$N_{\text{total}} = 1202$	Questionnaire:	
Refinement and validation of Scale	Exploratory Factor Analysis	$N_1 = 601$	FM-IDLES	EFA (SPSS 29.0)
	Confirmatory Factor Analysis	$N_2 = 601$		CFA (AMOS 26.0)
	Reliability and validity assessment			a). Reliability test: Cronbach's alpha, McDonald's ω b). Validity test: Convergent validity, discriminant validity (AVE, CR)
Study 2				
Investigation of the associations between FM-IDLE activities and L2 achievement and between FM-IDLE activities and foreign language confidence	Data collection for L2 achievement and foreign language confidence	$N_{\text{total}} = 1202$	- L2 proficiency is measured using the CET-4 score as a type of standardized test result. - The Foreign Language Confidence Scale - The FM-IDLES	- The reliability test - The validity test - The measurement model - The structural model (SPSS 29.0; AMOS 26.0)

Abbreviations: AVE = average variance extracted, CET-4 = National College English Test—Band 4, CFA = confirmatory factor analysis, CR = composite reliability, CV = convergent validity, EFA = exploratory factor analysis, FM-IDLES = form-focused informal digital learning of English and meaning-focused informal digital learning of English Scale.

communicators who could speak and use English confidently in power-loaded digital spaces. Building on these theoretical developments and empirical findings, we propose:

Hypothesis 3. *FIA has a positive influence on foreign language confidence.*

Hypothesis 4. *MIA has a positive influence on foreign language confidence.*

3 | Method

The present research comprises two interrelated studies addressing the proposed research questions. Study 1 focused on developing and validating a new instrument to measure Chinese EFL university students' involvement in FIA and MIA. Study 2 investigated the predictive relationships between these IDLE activities and learners' L2 achievement and foreign language confidence. The methodological details, including procedures, sampling, instrumentation, analytical tools, and data analysis approaches, are summarized in Table 1 and illustrated in Figure 1, with detailed descriptions provided in the subsequent sections.

3.1 | Research Context and Participants

This research comprised 1,202 university undergraduates enrolled in compulsory English courses across diverse geographical regions and institutions in China. Including

768 females (63.8%) and 434 males (36.2%), the participants represented a broad spectrum of academic disciplines, with STEM fields constituting the largest proportion (34.0%, $n = 409$), followed by Arts and Humanities (21.2%, $n = 255$), Social Sciences (16.8%, $n = 202$), Business and Management (9.5%, $n = 115$), and Health Sciences (2.6%, $n = 32$), with the remaining 15.7% ($n = 189$) distributed across other disciplines. The demographic composition of the sample reflected typical characteristics of Chinese undergraduate populations. The majority of participants (82.2%, $n = 989$) were aged between 18 and 20 years, with smaller proportions aged 20–22 (4.1%, $n = 50$) and above 22 (0.4%, $n = 5$). A notable minority (13.1%, $n = 158$) were under 18.

Participants' English achievement levels were assessed through multiple indicators. In terms of standardized test performance, 109 students (9.2%) achieved the benchmark score of 425 out of 710 on the National College English Test-4 (CET-4), while 28 students (2.1%) reached this threshold on the more advanced CET-6. The National College Entrance Examination (Gaokao) English scores indicated that the majority of participants (57.0%, $n = 686$) scored between 90–119 out of 150, with 23.6% ($n = 285$) scoring below 90 and 19.2% ($n = 231$) achieving scores above 120.

3.2 | Instruments

In addition to the self-developed MF-IDLES¹ (refer to Section 4 for detailed instrument information) (a 6-point Likert scale), the main instruments employed in the two sub-studies: a) Gaokao English scores, CET-4, and CET-6 scores; b) the Foreign Language

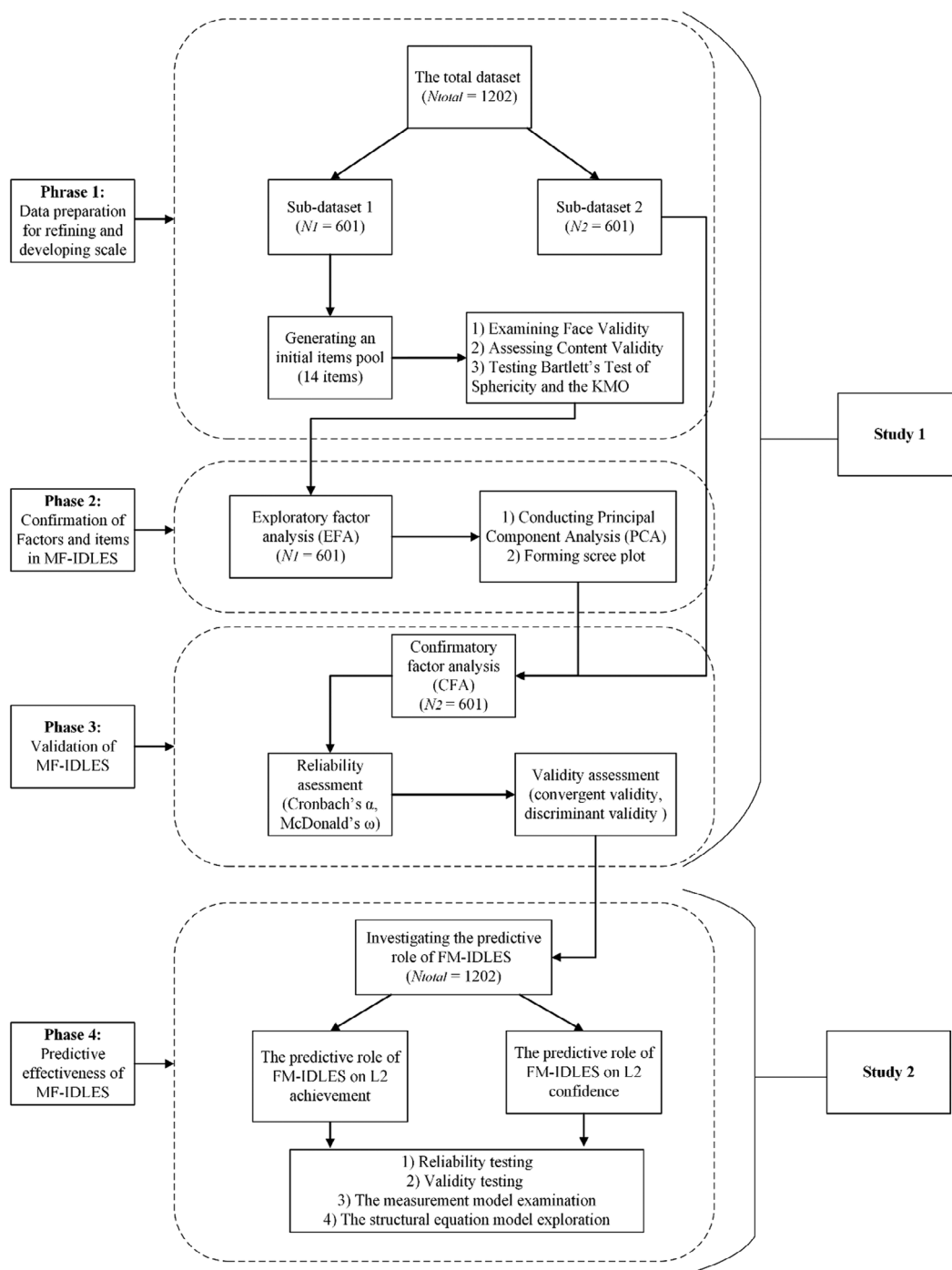


FIGURE 1 | Roadmap of research design procedures.

Confidence Scale (FLCS). All the scales were presented in Chinese. The items on the FLCS scale were responded to on a 6-point Likert scale ranging from 1 “Strongly disagree” to 6 “Strongly agree”. All instruments in the datasets demonstrated excellent reliability.

a. The standardized English assessments in the Chinese educational system align with the Common European Framework of Reference for Languages (Zhao et al. 2017). The National College Entrance Examination (Gaokao) English test has a maximum score of 150, while both the College English Test-4 (CET-4) and CET-6 have maximum scores of 710. As

Zhao et al. (2017) noted, for the CET-4, scores of 425–499 correspond to CEFR B1 level, 500–599 to B2 level, and scores above 600 indicate C1 level. The more advanced CET-6 sets higher benchmarks, with scores of 425–499 corresponding to B2 level and scores of 500 and above indicating C1 level. In the Gaokao English assessment, scores between 90 and 120 align with the CEFR B1 level, while scores of 121–150 indicate B2 level.

b. The FLCS containing 16 items was used to measure learners’ foreign language confidence from a World Englishes perspective (Liu 2024b), which has three dimensions: Foreign

Language Competence (FLC), Sense of Linguistic Security (SLS), Sense of Linguistic Ownership (SLO). FLC refers to learners' perceived assurance in their holistic ability to comprehend and use English for communicative and meaning-making purposes rather than focusing on discrete language skills. A sample item reads, "I feel confident in my ability to use English effectively in various situations." SLS measures learners' awareness of English language variation and their comfort in using diverse linguistic forms during English-mediated communication, both within and beyond their speech communities. This dimension is exemplified by the item "I feel at ease when speaking localized or accented English with others regardless of their cultural backgrounds." SLO assesses learners' perceived connection to, pride in, and identification with the English language, independent of their linguistic background. One example item: "I believe English is a language that belongs to all people who speak it."

3.3 | Data Collection and Ethics

We employed a convenience sampling technique to collect data through an online questionnaire accessible via QR code distribution to target participant groups. To ensure comprehension and response accuracy, all questionnaire items were presented in Chinese. The data collection process, conducted between September and November 2024, adhered to general research ethics guidelines and was administered by the fourth author following institutional approval. Prior to participation, all respondents were briefed on the study's purpose and provided informed consent.

4 | Findings of Study 1: Developing and Validating the FM-IDLE Scale

Study 1 focused on the refinement, development, and validation of the scale measuring students' involvement with form-focused IDLE activities and meaning-focused IDLE activities.

4.1 | Phase 1: Preparing the Data for Scale Refinement and Validation

To ensure robust psychometric validation, the complete dataset ($N_{\text{total}} = 1,202$) was randomly divided into two equal subsets ($N_1 = 601$, $N_2 = 601$). This split-sample approach is recommended in scale development research to avoid potential common method bias and to cross-validate the factor structure (Hair et al. 2019). The first dataset was employed for exploratory factor analysis (EFA), with the sample size ($N_1 = 601$) meeting the recommended criterion of ten participants per scale item (Hair et al. 2019). The second dataset was utilized for subsequent confirmatory factor analysis (CFA) with the sample size ($N_2 = 601$) to validate the factor structure that emerged from EFA.

4.2 | Phase 2: Developing the FM-IDLES and Confirming its Components

In Phase 2, a series of systematic procedures were implemented to develop the FM-IDLES and examine their underlying factorial

structure using the first dataset ($N_1 = 601$). The following stringent psychometric criteria guided the analysis.

4.2.1 | Item Generation and Analysis

Drawing upon existing IDLE literature (Lee 2020a, 2020b; Liu et al. 2024; Liu and Ma 2024; Zou et al. 2025c), an initial pool of 14 items was deductively generated to reflect the theoretical distinction between meaning-focused and form-focused informal learning activities. The scale underwent a two-stage initial validation process. First, face validity was examined through a pilot survey with 15 EFL undergraduates at Y University, who assessed the items' comprehensibility, clarity, unambiguity, and relevance for Chinese EFL learners. Based on their detailed feedback, several item statements were refined while maintaining the original 14-item structure. Subsequently, content validity was established through expert review. Five established quantitative researchers in applied English linguistics evaluated the items' pertinence, technical merit, and representativeness within their respective factors in the FM-IDLES. Following this content analysis, all 14 items were retained in the scale.

4.2.2 | EFA

To explore the underlying factor structure of FM-IDLES, EFA was conducted using SPSS 29.0. The data's suitability for factor analysis was confirmed through the Kaiser-Meyer-Olkin measure of sampling adequacy ($KMO = 0.95$, exceeding the critical value of 0.60) and Bartlett's test of sphericity ($p < 0.001$). The extraction methods were principal component analysis with the oblique (promax) rotation (Bro and Smilde 2014). Principal Component Analysis yielded a clear two-factor structure, with all items displaying satisfactory factor loadings and communalities (all > 0.40), supporting the retention of all items in the scale (see Table 2).

Factor retention decisions were also based on additional statistical criteria, including eigenvalues and the scree plot. Specifically, factors with eigenvalues exceeding one were identified for potential retention. Also, Cattell's scree plot was examined to identify the point of inflection where eigenvalues transition from a steep decline to a more gradual slope, with factors preceding this point considered for retention (Hair et al. 2019). The scree plot analysis (see Figure 2) provided additional support for the two-factor solution, with only two factors demonstrating eigenvalues exceeding unity (Factor 1 = 8.91, Factor 2 = 1.20). This pattern of eigenvalues strongly corroborated the theoretical conceptualization of FM-IDLES as a two-dimensional construct.

4.2.3 | CFA

CFA was conducted using Amos 26.0 with Dataset 2 ($N_2 = 601$) to validate the two-factor structure identified through EFA. The analysis employed maximum likelihood estimation and assessed model fit through multiple indices: Chi-square

TABLE 2 | Results of EFA.

Item No.	Factor 1	Factor 2	Communality	Cumulative % of variance (rotated)
F1. I actively listen to English news or podcasts on my mobile phone or computer outside class to improve my English listening skills.	0.66		0.65	36.42
F2. I actively use English learning apps to practice my oral expression and improve my English-speaking skills outside the classroom.	0.79		0.76	
F3. I actively use English learning apps to practice and improve my English pronunciation outside the classroom.	0.81		0.79	
F4. I actively improve my English reading skills by reading English articles/content through online resources outside the classroom.	0.72		0.74	
F5. I actively use mobile phone apps or websites for online English writing practice outside of class to improve my English writing skills.	0.63		0.76	
F6. I actively use vocabulary apps outside of class to expand my English vocabulary.	0.83		0.70	
F7. I actively use mobile phone apps or websites to learn English grammar outside of class.	0.72		0.71	
M1. I actively watch short English videos (e.g. vlogs) on social media outside the classroom.		0.52	0.63	72.21
M2. I actively communicate with other learners in English through social media apps outside the classroom.		0.82	0.79	
M3. I actively chat with foreigners in English through social media software outside the classroom.		0.89	0.84	
M4. I actively read English articles online or English-related posts on social media.		0.82	0.81	
M5. I actively participate in online English discussion platforms to discuss with others on topics that are currently popular or familiar to me.		0.87	0.84	
M6. I actively watch English-language films and TV series outside the classroom.		0.41	0.46	
M7. I actively search online (either in Chinese or English) to learn about the culture and history of English-speaking countries/languages to develop my intercultural communication skills.		0.67	0.64	
KMO (>0.60)	0.95			
Bartlett's test of sphericity	Approx. Chi-square (χ^2)	7348.67		
	Degree of freedom (df)	61		
	p	< 0.001		

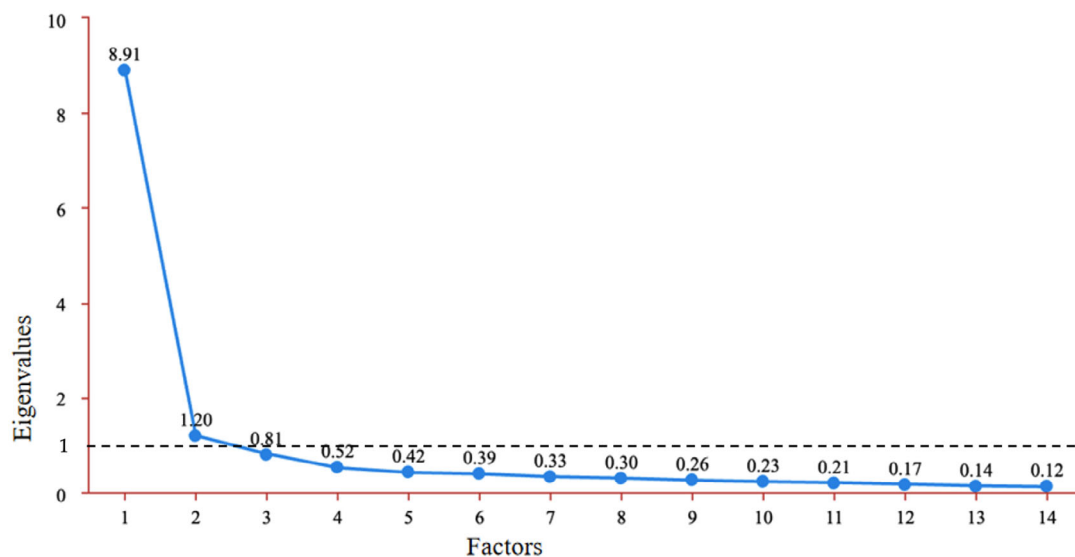


FIGURE 2 | Scree plot. [Color figure can be viewed at wileyonlinelibrary.com]

to Degrees of Freedom ratio (χ^2/df), Comparative Fit Index (CFI), Goodness-of-Fit Index (GFI), Incremental Fit Index (IFI), Root Mean Square Error of Approximation (RMSEA), Tucker–Lewis Index (TLI), and Standardized Root Mean Square Residual (SRMR) (Hu and Bentler 1999). The analysis included a thorough examination of factor loadings, cross-loadings, and residual variances.

There were two rounds of measurement model analysis conducted. In the first round, the results disclosed that the initial measurement model demonstrated poor fit with the dataset, as five out of the seven indices (i.e., χ^2/df , IFI, TLI, NFI, and CFI) did not reach the cut-off values. In this case, following the recommendation of Collier (2020), we refined the measurement model by calculating modification indices (MI) in Amos. Based on the MI value, two pairs of errors (i.e., eF2-eF3 and eM1-eM6) from the constructs FIA and MIA, respectively, were correlated to reduce item redundancy in the measurement of variables. The decision to correlate these errors was based on both statistical and theoretical considerations. Statistically, the MI values indicated that allowing these error terms to correlate would lead to a substantial improvement in the model fit. Theoretically, the correlated error terms were associated with items that exhibited similar content or wording within their respective constructs (i.e., FIA and MIA), a phenomenon referred to as method effects (Brown 2015). By allowing these errors to correlate, we aimed to improve the precision of our measurement model. After refining the initial measurement model, the final measurement model generally demonstrated a good fit with Dataset 2 (see Figure 3). While the χ^2/df (8.53) went beyond the recommended threshold of 5, possibly because of the sensitivity of Chi-square to sample size (Kline 2015), CFI (0.93), NFI (0.92), IFI (0.93), and TLI (0.91) values all exceeded the established cut-off criterion of 0.90 (Collier 2020). The PGFI value was higher than the established benchmark value of 0.50 (Collier 2020). Additionally, SRMR (0.06) values were below the 0.08 threshold recommended by Bowen and Guo (2011) (see Table 3). These results collectively confirmed the construct validity of the two-factor structure for the 14-item FM-IDLES.

4.3 | Phase 3: Validating the FM-IDLES

The newly developed 14-item FM-IDLES also underwent a comprehensive assessment of reliability and validity.

4.3.1 | Reliability Assessment

Following Hair et al. (2019), the reliability of FM-IDLES was evaluated through two measures of internal consistency: Cronbach's alpha (α) and McDonald's omega (ω). Both subscales demonstrated excellent reliability: form-focused IDLE activities ($\alpha = 0.93$, $\omega = 0.91$) and meaning-focused IDLE activities ($\alpha = 0.92$, $\omega = 0.90$). These robust reliability coefficients provide strong evidence for the internal consistency of the FM-IDLES.

4.3.2 | Validity Assessment

Convergent and discriminant validity of the FM-IDLES were examined through multiple criteria. Convergent validity was assessed via average variance extracted (AVE) and composite reliability (CR) scores. As shown in Table 4, both factors demonstrated satisfactory convergent validity, with AVE values exceeding the recommended threshold of 0.50 and CR values surpassing the cut-off criterion of 0.70 (Kline 2015). Discriminant validity was evaluated by comparing the square roots of AVE for each factor with their inter-factor correlation coefficients. The analysis revealed that the square roots of AVE for both factors exceeded their corresponding correlation coefficients, providing strong evidence for the distinct nature of the two factors.

5 | Findings of Study 2: The Associations between FM-IDLES and L2 Achievement and L2 Confidence

Study 2 was conducted to disclose the predictive effects of FIA and MIA on L2 English achievement and L2 confidence.

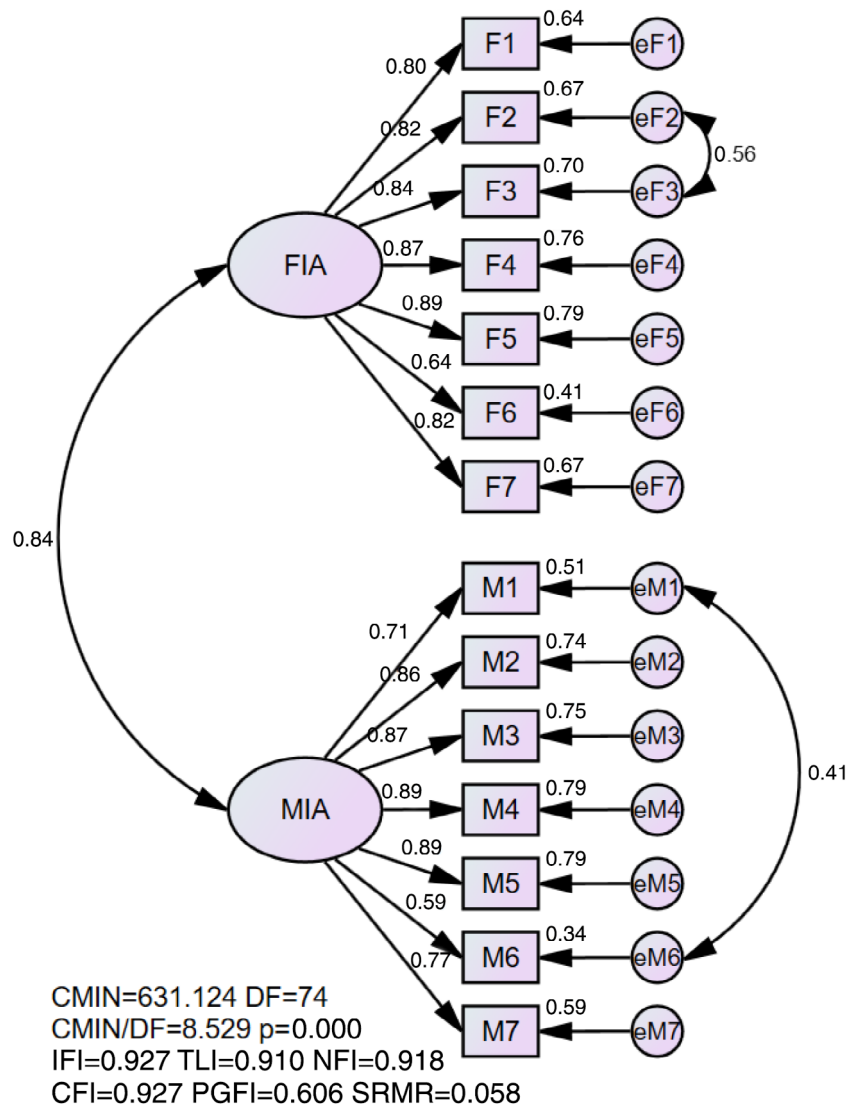


FIGURE 3 | The measurement model with standardized loadings. [Color figure can be viewed at wileyonlinelibrary.com]

TABLE 3 | Goodness model fit indices.

	χ^2/df	IFI	TLI	NFI	CFI	PGFI	SRMR
The first measurement model	11.92	0.89	0.87	0.88	0.89	0.58	0.06
The second measurement model	8.53	0.93	0.91	0.92	0.93	0.61	0.06
Benchmark value (Bowen and Guo 2011; Kline 2023)	<5	>0.90	>0.90	>0.90	>0.90	>0.50	<0.08

TABLE 4 | Convergent validity and discriminant validity.

Factor		α (>0.7)	Ω (>0.8)	AVE (>0.5)	CR (>0.7)	The square roots of AVE and squared correlation coefficient matrix	
						1	2
1	FIA	0.93	0.91	0.66	0.93	0.81	
2	MIA	0.92	0.90	0.63	0.92	0.79	0.79

TABLE 5 | Convergent and discriminant validity of FM-IDLES and L2 confidence.

	Factor	AVE (>0.5)	CR (>0.7)	The square roots of AVE and squared correlation coefficient matrix				
				1	2	3	4	5
1	FIA	0.68	0.94	0.82				
2	MIA	0.64	0.93	0.79	0.80			
3	Linguistic competence	0.75	0.94	0.50	0.65	0.86		
4	Linguistic security	0.72	0.94	0.42	0.60	0.84	0.85	
5	Language ownership	0.60	0.88	0.57	0.62	0.70	0.64	0.78

TABLE 6 | Goodness model fit indices.

	χ^2/df	IFI	TLI	NFI	CFI	PGFI	SRMR
The measurement model	7.01	0.92	0.91	0.904	0.92	0.71	0.06
The first structural model	7.41	0.91	0.904	0.898	0.91	0.71	0.08
The second structural model	6.88	0.92	0.91	0.906	0.92	0.71	0.06
Benchmark value (Kline, 2023)	<5	>0.90	>0.90	>0.90	>0.90	>0.50	<0.10

Note: The ratio " $\chi^2/df > 5$ " was presented due to the high sensitivity of χ^2 to variations in sample size.

5.1 | Data Analysis

An SEM approach (Collier 2020; Kline 2015) was employed to investigate the predictive validity of FM-IDLES using the complete dataset ($N_{\text{total}} = 1202$). The analysis proceeded sequentially, beginning with the estimation of a measurement model to validate the factor structure, followed by the specification of a structural model to examine inter-variable relationships through path analysis.

5.2 | Reliability and Validity Measurement

Using the complete sample, internal consistency reliability was assessed using Cronbach's alpha (α), with all variables demonstrating excellent reliability: FIA ($\alpha = 0.93$), MIA ($\alpha = 0.92$), and L2 Confidence ($\alpha = 0.96$, with subscales FLC = 0.94, SLS = 0.94, and SLO = 0.88). All coefficients exceeded the recommended threshold of 0.70 (Kline 2023), indicating strong internal consistency. The construct validity assessment encompassed both convergent and discriminant validity. As shown in Table 5, all fit indices met their respective recommended thresholds, providing robust evidence for the scale's construct validity.

5.3 | The Measurement Model and Structural Model Results

Following confirmation of the scales' excellent psychometric properties, the measurement model was tested using Amos. Despite the unstable χ^2/df , analysis of model fit indices generally demonstrated good alignment between the measurement model and the total dataset (see Table 6).

There were two rounds of structural model analysis conducted to test the hypotheses. In the first round, the results revealed that the proposed structural model exhibited poor fit with the dataset, as two out of the seven indices (i.e., χ^2/df , NFI) did not meet the established benchmark values. In this instance, we adhered to the recommendation of Collier (2020) by calculating modification indices (MI) in Amos to refine the model. Building on the MI value, one pair of errors (i.e., eF2–eF3) from the factor FIA was correlated, with the aim of reducing item redundancy in the measurement of variables. After doing so, the second structural model displayed a good fit with the dataset because all model indices fell in the recommended indices (see Figure 4).

Based on the second structural model, hypotheses measurement results were summarized in Table 7, indicating that four hypotheses were supported. Specifically, FIA played a negative and significant predictive role in actual L2 English achievement² (H1: $\beta = -0.16$, $p = 0.014$, $t\text{-value} = -2.46$). MIA enabled to predict actual L2 English achievement positively and significantly (H2: $\beta = 0.19$, $p = 0.004$, $t\text{-value} = 2.88$). FIA was able to predict L2 confidence in a negative and significant way (H3: $\beta = -0.19$, $p < 0.001$, $t\text{-value} = -3.79$), while MIA can make a positive and significant impact on L2 confidence (H4: $\beta = 0.90$, $p < 0.001$, $t\text{-value} = 15.87$). Examination of R^2 values (see Figure 4) revealed differential predictive power across outcomes: FIA and MIA collectively explained only 1% of the variance in L2 English Achievement, suggesting the influence of other unmeasured variables in determining achievement levels. In contrast, these predictors accounted for a substantial 56% of the variance in L2 confidence, indicating that FM-IDLE activities play a particularly crucial role in shaping learners' confidence levels. The elevated R^2 value indicated the remarkable explanatory capacity of the proposed model, potentially implying that four of the significant hypotheses exhibited substantial effect sizes (Plonsky and Oswald 2014).

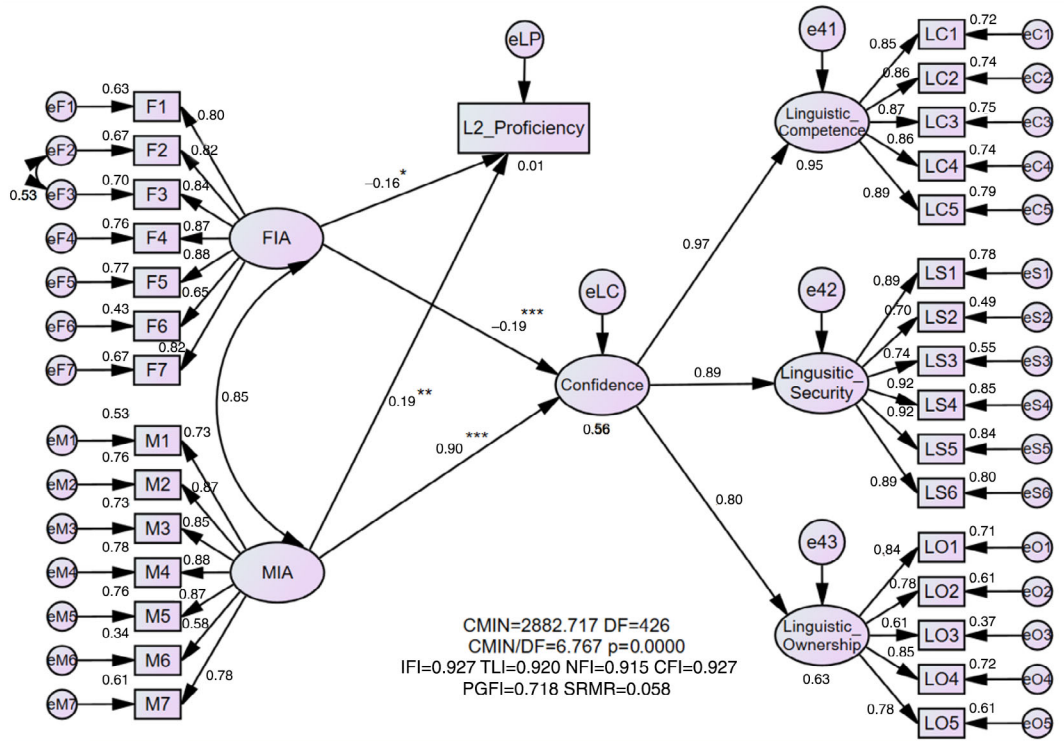


FIGURE 4 | The structural model of MF-IDLE, L2 achievement, and L2 confidence. $p < 0.001^{***}$, $p < 0.01^{**}$, $p < 0.05^{*}$. **Abbreviations:** FIA = form-focused informal digital learning of English activities, MIA = meaning-focused informal digital learning of English activities; (R^2 : L2 Achievement = 1%, L2 Confidence = 56%). [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com)]

TABLE 7 | Hypotheses test results.

Hypotheses	β	S.E.	C.R. (t -value) (> 1.96)	p	Results
H1: FIA $\rightarrow$ L2 Achievement	-0.16*	0.05	-2.46	0.014	Supported
H2: MIA $\rightarrow$ L2 Achievement	0.19**	0.05	2.88	0.004	Supported
H3: FIA $\rightarrow$ L2 Confidence	-0.19***	0.06	-3.79	<0.001	Supported
H4: MIA $\rightarrow$ L2 Confidence	0.90***	0.06	15.87	<0.001	Supported

Note: $p < 0.001^{***}$, $p < 0.01^{**}$, $p < 0.05^{*}$.

6 | Discussion

6.1 | Psychometric Properties of the FM-IDLES

The first research question examined the reliability and validity of FM-IDLES as a first-order, two-factor scale. Our findings provide comprehensive evidence supporting the scale's psychometric properties. Results of EFA clearly identified a two-factor structure, with all items demonstrating satisfactory factor loadings on their respective dimension. CFA subsequently confirmed this factorial structure, which demonstrated good model fit indices and helped statistically validate the theoretical distinction between form-focused and meaning-focused IDLE activities. The scale's psychometric robustness was further established through multiple reliability and validity checks. Both dimensions demonstrated excellent internal consistency, as evidenced by strong reliability coefficients (Cronbach's α and McDonald's ω). The scale also exhibited satisfactory convergent and discriminant

validity, confirming that FIA and MIA represent distinct yet related constructs.

Building on these findings, we argue that the validated two-factor structure of FM-IDLES enables a more nuanced measurement of informal digital learning activities among EFL learners. While prior instruments have generally conceptualized IDLE as a unitary construct encompassing both receptive and productive engagement (Ghasemi and Azari Noughabi 2024; Lee and Drajeti 2019; Zadorozhnyy and Lee 2023), the FM-IDLES delineates between activities oriented toward linguistic form and those directed at authentic communicative use. This differentiation is consistent with theoretical models of second language development that highlight the distinct cognitive mechanisms underpinning form-focused and meaning-focused learning processes (Ellis 2001; Svalberg 2021), and it addresses scholarly calls for more nuanced assessments of out-of-class digital English learning practices (Reinders and Benson 2017).

While these findings establish the initial validity of the FM-IDLES as a measurement instrument, further validation studies are needed to strengthen its applicability across different contexts. Future research should focus on cross-validating the FM-IDLES across different L2 contexts and populations. To this end, following Collier's (2020) recommendations, multi-group CFA should be conducted to verify measurement invariance across demographic variables such as age, gender, academic major, and proficiency level. Additionally, longitudinal validation studies could examine the scale's stability over time, addressing the dynamic nature of informal English learning practices (Chik 2019; Liu 2024a; Sundqvist 2024).

6.2 | Contributions of FM-IDLE Activities to L2 English Achievement

The second research question investigated how FIA and MIA predict L2 English achievement, as measured by performance on the College English Test (CET-4), a high-stakes standardized test in China. Our findings reveal contrasting effects of these two types of IDLE activities on learners' test performance, though notably, their combined explanatory power was limited, accounting for only 1% of the variance in L2 achievement scores.

While FIA demonstrated a negative predictive relationship with L2 achievement ($\beta = -0.16$, $p = 0.014$, $t\text{-value} = -2.46$), rejecting Hypothesis 1 and diverging from previous findings (Lee and Dressman 2018; Sundqvist and Uztosun 2024), the small effect size suggests this relationship should be interpreted cautiously. Several factors may explain this weak negative association. First, excessive focus on linguistic forms in informal digital contexts may come at the expense of developing practical communicative abilities. Second, form-focused digital activities often lack authentic contexts and meaningful interactions, which potentially hinders the internalization of linguistic knowledge. Third, while FIA may contribute to the attainment of specific language knowledge, these improvements might not be adequately captured by the standardized achievement test used in our study (i.e., CET-4). The format and requirements of CET-4, which emphasize formal linguistic knowledge and test-taking strategies, may not fully capture the linguistic competencies acquired through informal digital English learning activities. This could suggest a need for more fine-grained evaluations that are attuned to the specific linguistic components targeted by FIA. Last, it is worth considering whether FIA might contribute more significantly to other dimensions of language learning that are not directly assessed by the CET-4, such as motivation, self-efficacy, or engagement. While these factors have not been explicitly measured in our study, they may serve pivotal functions in learners' long-term language development. As such, we call for further investigation.

Similarly, while MIA showed a positive predictive relationship with L2 achievement ($\beta = 0.19$, $p = 0.004$, $t\text{-value} = 2.88$), supporting Hypothesis 2 and aligning with previous research (Lee and Dressman 2018; Lai et al. 2015), the small effect size indicates that meaning-focused digital activities alone may not be sufficient to substantially improve standardized test performance. This finding suggests that while authentic language use and meaningful interaction in digital environments may contribute

positively to L2 development (Liu and Darvin 2024), their impact on formal assessment outcomes might be moderated or mediated by other unmeasured variables. This is particularly relevant given that standardized tests like CET-4 may not fully reflect the communicative competencies developed through meaning-focused digital activities. In addition, L2 achievement development is a complex phenomenon influenced by numerous factors beyond informal digital learning activities (De Wilde et al. 2022). Test performance is shaped by multiple variables, including formal instruction, individual differences in motivation and aptitude, exposure to the target language, and specific test preparation strategies (Ellis 2001). Therefore, while IDLE activities may play a role in enhancing students' overall language development, their limited relationship with CET-4 scores suggests the need for identifying more relevant variables and developing multiple measures of L2 achievement that can better capture the diverse linguistic competencies developed through both form-focused and meaning-focused digital learning activities in out-of-class environments.

6.3 | Predictive Effects of FM-IDLE Activities on L2 Confidence

The third research question examined how FIA and MIA predict L2 confidence, conceptualized through a World Englishes perspective that includes learners' linguistic competence, sense of linguistic security, and sense of linguistic ownership (Liu 2024b). Our findings reveal substantial differential effects of these IDLE activities on L2 confidence, collectively explaining 56% of its variance.

FIA demonstrated a significant negative relationship with L2 confidence ($\beta = -0.19$, $p < 0.001$, $t\text{-value} = -3.79$), rejecting Hypothesis 3, which diverges from Lee and Draji's (2019) findings. This negative association can be understood through several mechanisms. First, excessive attention to linguistic forms may generate anxiety about accuracy, undermining learners' confidence not only in their linguistic competence but also in their legitimacy as English users in a globalized context (Liu 2024b). Second, the limited interactive nature of form-focused activities may restrict opportunities for authentic feedback and cultural exchange, potentially hampering learners' development of linguistic security and awareness of language variation. Third, the decontextualized nature of many form-focused activities may limit learners' opportunities to develop a sense of ownership over English as a global language (Lai et al. 2015).

In contrast, MIA showed a remarkably strong positive relationship with L2 confidence ($\beta = 0.90$, $p < 0.001$, $t\text{-value} = 15.87$), supporting Hypothesis 4. This robust finding aligns with theoretical perspectives on how meaningful engagement with English in digital spaces can foster confidence through authentic intercultural experiences (Lee and Lee 2020; Liu and Darvin 2024; Liu et al. 2023). The strong predictive power of MIA suggests that meaning-focused digital activities particularly support the development of L2 confidence by potentially strengthening learners' belief in their communicative abilities, enhancing their appreciation of language variation, and fostering their sense of legitimate ownership of English as a global language (Liu 2024b). This relationship may be especially powerful because

meaning-focused activities often involve authentic intercultural communication, promoting learners' autonomy and self-directed learning while simultaneously validating their status as legitimate users of English in global digital spaces (Lamb and Arisandy 2020; Lee et al. 2021). Thus, the strong explanatory power of IDLE suggests that informal digital learning experiences, particularly meaning-focused ones, play a central role in developing confident, legitimate users of English in today's globalized and digital world.

6.4 | Pedagogical Implications

The findings of this study yield several significant pedagogical implications. First, the FM-IDLES can serve as a diagnostic tool for language educators to evaluate their students' participation in FIA and MIA categories of IDLE. By identifying students' inclinations and patterns of engagement with FIA and MIA activities, teachers can provide targeted teaching interventions. For instance, if a student exhibits a strong propensity towards MIA rather than FIA participation, the teacher may provide form-focused teaching tasks or guidance to assist the student in achieving a more balanced learning approach. Second, by comprehending the specific linguistic benefits associated with FIA and MIA activities, educators can make more informed choices regarding the digital resources and content they utilize to support language learning. This may involve curating a blend of form-focused and meaning-focused resources that align with students' language competence, learning goals, and interests. Given the differential effects of FIA and MIA on both L2 achievement and L2 confidence, educators should facilitate a more balanced approach to integrating IDLE into the classroom. Specifically, while form-focused activities need not be abandoned, they should be more strategically integrated with meaningful L2 communication in online and out-of-class environments. For instance, teachers could guide students in using language learning applications like Duolingo or Grammarly alongside participation in online discussion forums, ensuring that grammar and vocabulary learning are contextualized within authentic communication tasks. This integration could be achieved through structured digital tasks such as collaborative online projects where students first study specific linguistic features (e.g., academic vocabulary or discourse markers) and then apply them in real-world tasks like creating shared digital content or participating in international online debates. The careful selection of digital tools that provide immediate feedback within meaningful contexts is also crucial. One example practice is using AI writing assistants that not only highlight language errors but also suggest improvements while students engage in authentic writing tasks like blogging or social media communication. Additionally, structured reflection activities, such as weekly digital learning journals or group discussions about different IDLE experiences, could help students develop metacognitive awareness of how various digital learning activities contribute to their language development.

In addition, the strong positive relationship between MIA and L2 confidence ($R^2 = 56\%$) suggests the need for systematic institutional support of meaning-focused L2 practices in out-of-class online spaces. Educational institutions could facilitate virtual exchange programs with international partners through platforms like Zoom or Microsoft Teams, where students engage

in regular themed discussions or collaborative projects with peers from different cultural backgrounds. Universities could establish online communities of practice through social media groups or discussion forums where students engage in meaningful English communication around shared interests such as global issues, popular culture, or academic disciplines. For example, institutions might create moderated WeChat groups for different interest areas, encouraging students to share resources, discuss topics, and organize online events in English (Zou et al. 2025a). Such initiatives would provide authentic contexts for developing not only linguistic competence but also students' sense of legitimacy as English users in digital spaces. Moreover, institutions could organize regular online cultural exchange events, virtual book clubs, or student-led webinars where learners can practice English in meaningful contexts while building confidence in their ability to communicate across cultural boundaries.

7 | Conclusion

This study has developed and validated FM-IDLES by reporting its robust psychometric properties in measuring Chinese university EFL learners' involvement with informal digital learning practices. Additionally, we investigated the predictive relationships between different types of IDLE activities and both L2 achievement and L2 confidence. The findings revealed differential effects: while meaning-focused IDLE activities showed positive associations with both outcomes, form-focused activities demonstrated unexpected negative relationships.

Despite these important findings, several limitations of this study suggest directions for future research. First, while our convenience sampling through Chinese social media platforms provided access to public university students, future studies should include participants from private institutions to achieve more representative sampling. Second, the cross-sectional nature of our study cannot capture the dynamic patterns of IDLE involvement over time, thus suggesting the need for longitudinal investigations to understand how learners' digital learning practices evolve. The cross-sectional design limits causal inferences, and self-reported data may introduce bias, thereby recommending the need for experimental studies to explore the mechanism of causal inferences. Third, the application of FM-IDLES in a single EFL context (i.e., China) calls for diverse educational contexts (e.g., primary or secondary education) and cultural settings (e.g., non-Asian EFL learners) to establish measurement invariance across groups and broader applicability. Additionally, future studies might consider employing mixed-methods approaches to better understand the mechanisms through which form-focused and meaning-focused digital activities influence various aspects of language development and learner psychology. [Supplementary Material](#).

Acknowledgments

We are thankful to all participants involved in this research. Prior to their participation, we received their written consents. We also very much appreciate the editor and peer reviewers' valuable and insightful feedback. For the purpose of open access, Minlin Zou (affiliated University of Exeter) has applied a Creative Commons Attribution (CC

BY) licence to any Minlin Zou (affiliated Univeristy of Exeter) Accepted Manuscript version arising from this submission. ChatGPT did not contribute to the conception of any ideas presented in this article. In addition, we appreciate Hunan University of Science and Technology 2023 College Students' Innovation and Entrepreneurship Training Program Project in China (Project No.: S202310534118): *A Comparative Study on Metadiscourse in Commencement Speeches by Chinese and American University Presidents* (to Linghua Zhou).

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The raw data can be made available upon request.

Endnotes

¹ Please refer to the appendix for detailed information about the MF-IDLES items.

² In this study, L2 English achievement was assessed using their CET-4 scores because all participants have sat this exam and indicated their CET-4 score ranges in the questionnaire data.

References

- Benson, P. 2011. "Language Learning and Teaching Beyond the Classroom: An Introduction to the Field." In *Beyond the Language Classroom*, edited by P. Benson and H. Reinders, 7–16. Palgrave Macmillan. <https://doi.org/10.1057/97802303067902>.
- Bowen, N. K., and S. Guo. 2011. *Structural Equation Modeling*. Oxford University Press.
- Bro, R., and A. K. Smilde. 2014. "Principal Component Analysis." *Analytical Methods* 6, no. 9: 2812–2831. <https://doi.org/10.1039/C3AY41907J>.
- Brown, T. A. 2015. *Confirmatory Factor Analysis for Applied Research*, 2nd ed. Guilford Press.
- Chik, A. 2019. "Motivation and Informal Language learning." In *The Handbook of Informal Language Learning*, edited by M. Dressman and R. W. Sadler, 13–26. John Wiley & Sons Ltd. <https://doi.org/10.1002/9781119472384.ch1>.
- Collier, J. E. 2020. *Applied Structural Equation Modeling Using AMOS: Basic to Advanced Techniques*. Routledge.
- De Wilde, V., M. Brysbaert, and J. Eyckmans. 2022. "Formal Versus Informal L2 Learning: How Do Individual Differences and Word-Related Variables Influence French and English L2 Vocabulary Learning in Dutch-Speaking Children?." *Studies in Second Language Acquisition* 44, no. 1: 87–111. <https://doi.org/10.1017/S027226312100009>.
- Ellis, N. C., and S. Wulff. 2020. "Usage-based Approaches to L2 acquisition." In *Theories in Second Language Acquisition: An Introduction*, edited by B. VanPatten, G. D. Keating, and S. Wulf, 3rd ed., 63–82. Routledge.
- Ellis, R. 2001. "Introduction: Investigating Form-focused Instruction." *Language Learning* 51: 1–46. <https://doi.org/10.1111/j.1467-1770.2001.tb00013.x>.
- Ghasemi, A., and M. Azari Noughabi. 2024. "Investigating the Mediating Role of Foreign Language Peace of Mind and Learning Burnout in the Relationship Between Informal Digital Learning of English and Willingness to Communicate." *Computer Assisted Language Learning* 0, no. 0: 1–30. <https://doi.org/10.1080/09588221.2024.2424321>.
- Hair, J. F., W. C. Black, B. J. Babin, and R. E. Anderson. 2019. *Multivariate Data Analysis*, 8th ed. Cengage.
- Hu, L. T., and P. M. Bentler. 1999. "Cutoff Criteria for Fit Indexes in Covariance Structure Analysis: Conventional Criteria versus New Alternatives." *Structural Equation Modelling* 6, no. 1: 1–55. <https://doi.org/10.1080/10705519909540118>.
- Jones, R. H., and C. A. Hafner. 2021. *Understanding Digital Literacies: A Practical Introduction*. Routledge.
- Kaatari, H., T. Larsson, Y. Wang, S. Acikara-Eickhoff, and P. Sundqvist. 2023. "Exploring the Effects of Target-Language Extramural Activities on Students' Written Production." *Journal of Second Language Writing* 62: 101062. <https://doi.org/10.1016/j.jslw.2023.101062>.
- Kline, R. B. 2015. *Principles and Practice of Structural Equation Modeling*. Guilford Publications.
- Kline, R. B. 2023. *Principles and Practice of Structural Equation Modelling* (5th ed.). Guilford publications.
- Lai, C., and Q. Wang. 2024. "Online Informal Learning of English and Receptive Vocabulary Knowledge: Purpose Matters." *ReCALL* 1–17. <https://doi.org/10.1017/S095834402400017X>.
- Lai, C., W. Zhu, and G. Gong. 2015. "Understanding the Quality of out-of-class English Learning." *TESOL Quarterly* 49, no. 2: 278–308. <https://doi.org/10.1002/tesq.171>.
- Lamb, M., and F. E. Arisandy. 2020. "The Impact of Online Use of English on Motivation to Learn." *Computer Assisted Language Learning* 33, no. 1–2: 85–108. <https://doi.org/10.1080/09588221.2018.1545670>.
- Lantolf, J. P., and S. L. Thorne. 2006. *Sociocultural Theory and the Genesis of Second Language Development*. Oxford University Press.
- Lee, J. S. 2019. "Informal Digital Learning of English and Second Language Vocabulary Outcomes: Can Quantity Conquer Quality?" *British Journal of Educational Technology* 50, no. 2: 767–778. <https://doi.org/10.1111/bjet.12599>.
- Lee, J. S. 2020a. "Informal Digital Learning of English and Strategic Competence for Cross-cultural Communication: Perception of Varieties of English as a Mediator." *ReCALL* 32, no. 1: 47–62. <https://doi.org/10.1017/S0958344019000181>.
- Lee, J. S. 2020b. "The Role of Informal Digital Learning of English and a High-Stakes English Test on Perceptions of English as an International Language." *Australasian Journal of Educational Technology* 36, no. 2: 155–168. <https://doi.org/10.14742/ajet.5319>.
- Lee, J. S., and N. A. Drajiati. 2019. "Affective Variables and Informal Digital Learning of English: Keys to Willingness to Communicate in a Second Language." *Australasian Journal of Educational Technology* 35, no. 5: 168–182. <https://doi.org/10.14742/ajet.5177>.
- Lee, J. S., and M. Dressman. 2018. "When IDLE Hands Make an English Workshop: Informal Digital Learning of English and Language Proficiency." *TESOL Quarterly* 52, no. 2: 435–445. <https://doi.org/10.1002/tesq.422>.
- Lee, J. S., and K. Lee. 2020. "Affective Factors, Virtual Intercultural Experiences, and L2 Willingness to Communicate in In-Class, Out-of-Class, and Digital Settings." *Language Teaching Research* 24, no. 6: 813–833. <https://doi.org/10.1177/1362168819831408>.
- Lee, J. S., and Y. Lu. 2023. "L2 motivational Self System and Willingness to Communicate in the Classroom and Extramural Digital Contexts." *Computer Assisted Language Learning* 36, no. 1–2: 126–148. <https://www.tandfonline.com/doi/abs/10.1080/09588221.2021.1901746>.
- Lee, J. S., and T. Taylor. 2024. "Positive Psychology Constructs and Extramural English as Predictors of Primary School Students' willingness to Communicate." *Journal of Multilingual and Multicultural Development* 45, no. 7: 2898–2916. <https://doi.org/10.1080/01434632.2022.2079650>.
- Lee, J. S., Q. Xie, and K. Lee. 2021. "Informal Digital Learning of English and L2 Willingness to Communicate: Roles of Emotions, Gender, and Educational Stage." *Journal of Multilingual and Multicultural Development* 1–17. <https://doi.org/10.1080/01434632.2021.1918699>.
- Liu, G., and R. Darwin. 2024. "From Rural China to the Digital Wilds: Negotiating Digital Repertoires to Claim the Right to Speak." *TESOL Quarterly* 58, no. 1: 334–362. <https://doi.org/10.1002/tesq.3233>.
- Liu, G., R. Darwin, and C. Ma. 2024a. "Exploring AI-mediated Informal Digital Learning of English (AI-IDLE): A Mixed-Method Investigation of

- Chinese EFL Learners' AI Adoption and Experiences." *Computer Assisted Language Learning* 1–29. <https://doi.org/10.1080/09588221.2024.2310288>.
- Liu, G., and C. Ma. 2024. "Measuring EFL Learners' use of ChatGPT in Informal Digital Learning of English Based on the Technology Acceptance Model." *Innovation in Language Learning and Teaching* 18, no. 2: 125–138. <https://doi.org/10.1080/17501229.2023.2240316>.
- Liu, G., C. Ma, J. Bao, and Z. Liu. 2023. "Toward a Model of Informal Digital Learning of English and Intercultural Competence: A Large-Scale Structural Equation Modeling Approach." *Computer Assisted Language Learning* 1–25. <https://doi.org/10.1080/09588221.2023.2191652>.
- Liu, G. L. 2024a. "Modeling L2 Motivation Change and Its Predictive Effects on Learning Behaviors in the Extramural Digital Context: A Quantitative Investigation in China." *Linguistics Vanguard* 1–12. <https://doi.org/10.1515/lingvan-2023-0145>.
- Liu, G. L. 2024b. "Revisiting and Measuring Foreign Language Confidence From a World Englishes Perspective: Scale Development and Validation." *International Journal of Applied Linguistics* 1–17. <https://doi.org/10.1111/ijal.12618>.
- Liu, G. L. 2025. "Digital Empowerment for Rural Migrant Students in China: Identity, Investment, and Digital Literacies Beyond the Classroom." *ReCALL* 1–17. <https://doi.org/10.1017/S0958344024000302>.
- Liu, G. L., R. Darwin, and C. Ma. 2024b. "Unpacking the Role of Motivation and Enjoyment in AI-mediated Informal Digital Learning of English (AI-IDLE): A Mixed-Method Investigation in the Chinese Context." *Computers in Human Behavior* 160: 108362. <https://doi.org/10.1016/j.chb.2024.108362>.
- Liu, G. L., Y. Zhang, and R. Zhang. 2024. "Examining the Relationships Among Motivation, Informal Digital Learning of English, and Foreign Language Enjoyment: An Explanatory Mixed-Method Study." *ReCALL* 36, no. 1: 72–88. <https://doi.org/10.1017/S0958344023000204>.
- Liu, G. L., M. M. Zou, A. Soyoof, and M. M. Chiu. 2024. "Untangling the Relationship Between AI-Mediated Informal Digital Learning of English (AI-IDLE), Foreign Language Enjoyment and the Ideal L2 self: Evidence From Chinese University EFL Students." *European Journal of Education* 60, no. 1 e12846. <https://doi.org/10.1111/ejed.12846>.
- Peters, E., A.-S. Noreillie, K. Heylen, B. Bulté, and P. Desmet. 2019. "The Impact of Instruction and Out-of-School Exposure to Foreign Language Input on Learners' Vocabulary Knowledge in Two Languages." *Language Learning* 69, no. 3: 747–782. <https://doi.org/10.1111/lang.12351>.
- Plonsky, L., and F. L. Oswald. 2014. "How Big Is 'Big'? Interpreting Effect Sizes in L2 Research." *Language Learning* 64, no. 4: 878–912. <https://doi.org/10.1111/lang.12079>.
- Reinders, H., and P. Benson. 2017. "Research Agenda: Language Learning Beyond the Classroom." *Language Teaching* 50, no. 4: 561–578. <https://doi.org/10.1017/S0261444817000192>.
- Rezai, A., A. Soyoof, and B. L. Reynolds. 2024. "Effectiveness of Informal Digital Learning of English on EFL Learners' Vocabulary Knowledge: A Mixed-Methods Investigation." *Computer Assisted Language Learning* 1–26. <https://doi.org/10.1080/09588221.2024.2350419>.
- Skehan, P. 1989. *Individual Differences in Second-language Learning*. Edward Arnold.
- Skehan, P. 1991. "Individual Differences in Second Language Acquisition." *Studies in Second Language Acquisition* 13, no. 2: 275–298. <https://doi.org/10.1017/S0272263100009979>.
- Soyoof, A., B. L. Reynolds, B. Vazquez-Calvo, and K. McLay. 2023. "Informal Digital Learning of English (IDLE): A Scoping Review of What Has Been Done and a Look Towards What is to Come." *Computer Assisted Language Learning* 36, no. 4: 608–640. <https://doi.org/10.1080/09588221.2021.1936562>.
- Sundqvist, P. 2024. "Extramural English as an Individual Difference Variable in L2 Research: Methodology Matters." *Annual Review of Applied Linguistics* 1–13. <https://doi.org/10.1017/S0267190524000072>.
- Sundqvist, P., and M. S. Uztosun. 2024. "Extramural English in Scandinavia and Asia: Scale Development, Learner Engagement, and Perceived Speaking Ability." *TESOL Quarterly* 58, no. 4: 1638–1665. <https://doi.org/10.1002/tesq.3296>.
- Svalberg, A. M. 2021. "Engagement With Language in Relation to Form-Focused Versus Meaning-Focused Teaching and Learning." In *Student Engagement in the Language Classroom*, edited by P. Hiver, A. H. Al-Hoorie, and S. Mercer, 38–55. De Gruyter.
- Thorne, S. L. 2014. "Mediating Technologies and Second Language Learning." In *Handbook of Research on New Literacies*, edited by J. Coiro, M. Knobel, C. Lankshear, and D. J. Leu, 415–448. Routledge. <https://doi.org/10.4324/9781410618894>.
- Zadorozhnyy, A., and J. S. Lee. 2023. "Informal Digital Learning of English and Willingness to Communicate in a Second Language: Self-Efficacy Beliefs as a Mediator." *Computer Assisted Language Learning* 1–21. <https://doi.org/10.1080/09588221.2023.2215279>.
- Zhang, Y., and G. Liu. 2022. "Revisiting Informal Digital Learning of English (IDLE): A Structural Equation Modeling Approach in a University EFL Context." *Computer Assisted Language Learning* 1–33. <https://doi.org/10.1080/09588221.2022.2134424>.
- Zhao, W., B. Wang, D. Coniam, and B. Xie. 2017. "Calibrating the CEFR Against the China Standards of English for College English Vocabulary Education in China." *Language Testing in Asia* 7: 1–18. <https://doi.org/10.1186/s40468-017-0036-1>.
- Zou, M., M. Azari Noughabi, and C. Peng. 2025a. "Modeling the Associations Between L2 Grit, Foreign Language Enjoyment, and Student engagement Among Chinese EFL English-Major Learners: A control-Value the Oryperspective." *System* 131, no. 103679: 1–14. <https://doi.org/10.1016/j.system.2025.103679>.
- Zou, M., M. Azari Noughabi, M. Sabouri, and L. Zhou. 2025b. "The Contribution of Informal Digital Learning of English (IDLE) To achievement Emotions and Willingness to Communicate: A Cross-Cultural Investigation." *Journal of Multilingual and Multicultural Development* 1–19. <https://doi.org/10.1080/01434632.2025.2490076>.
- Zou, M. M., G. L. Liu, D. Li, and H. Chen. 2025c. "Beyond Motivation: Modeling the Predictive Role of L2 Pride in Informal Digital Learning of English." *International Journal of Applied Linguistics* 1–15. <https://doi.org/10.1111/ijal.12738>.

Supporting Information

Additional supporting information can be found online in the Supporting Information section.